GitHub "About" box (repository settings)

Description (copy into the About box): Mobile Price Classification: can PCA reduce 20 phone features while preserving important information? Python and Google Colab ML mini-project.

Topics (add these tags): `machine-learning` `pca` `dimensionality-reduction` `python` `scikit-learn` `google-colab` `data-science` `eda` `jupyter-notebook` `mobile-price-classification`

Website (optional): paste your Colab link, for example `https://colab.research.google.com/github/YOUR_USERNAME/YOUR_REPO/blob/main/PCA_Mobile_Project_Colab.ipynb`

Short project summary (for the top of the repo or a report)

This project investigates whether Principal Component Analysis (PCA) can reduce the 20 features of the Kaggle Mobile Price Classification dataset while keeping the information needed to predict the price range. After cleaning and scaling the data, the project explores feature correlations, applies PCA, and compares a logistic regression classifier trained on all features with one trained on PCA components. It finds that PCA gives only a small reduction (about 17 of 20 components are needed to keep accuracy), because most features are uncorrelated and the key feature, `ram`, is independent of the others.